



MICROBOT ROVER V1.3

Documentation

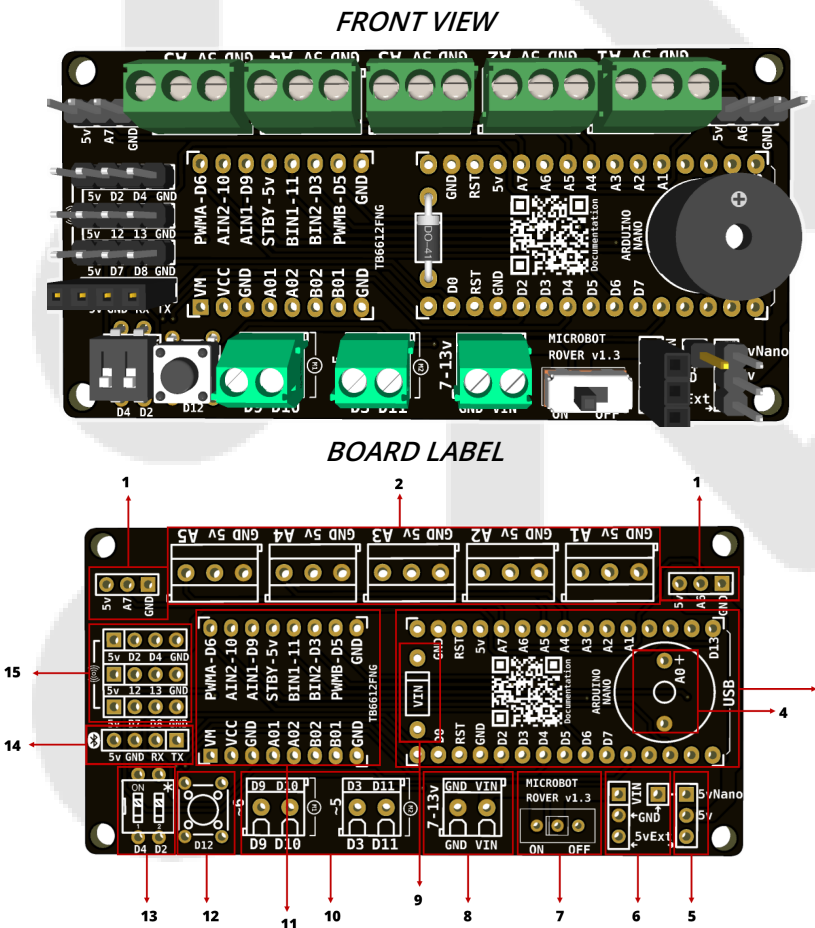
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Overview:

Microbot is a versatile expansion board designed for the **Arduino Nano** microcontroller, tailored to empower beginners and intermediate-level makers, tinkerers, and robotics enthusiasts. It serves as a robust foundation for building mobile projects, combining ease of use with powerful functionality.

At its core, **Microbot** integrates the reliable **TB6612FNG** dual motor driver, enabling precise control of DC motors for robotics and automation tasks. With its thoughtfully arranged layout, the board offers effortless connectivity to a wide range of sensors, modules, and external power supplies—making prototyping faster and more efficient.

Board Layout:



Key Features:

- 3 Ultrasonic sensor connections
- 7 IR sensor connections (terminal blocks and male headers)
- 1 Wireless communication module connection (Bluetooth or similar)
- 2 DIP switches (connected to D4 and D2)
- 1 Tactile switch (connected to D12)
- Arduino Nano mounting slot
- TB6612FNG motor driver mounting slot
- Active buzzer mount
- Main power ON/OFF switch
- External power input terminal
- Power source selection jumper (Internal / External)

Pin Layout Documentation		
Label	Pin	Description
1, 2	A1, A2, A3, A4, A5, A6, A7	Connected via a 3-pin terminal block. This setup is ideal for 3-pin sensors, such as IR sensors, SHARP sensors, and similar devices.
11	Motor Driver Slot	This header pin is used to connect the TB6612FNG module—a highly efficient, MOSFET-based dual DC motor driver IC capable of controlling two DC motors or a single stepper motor. It supports a continuous current of 1.2A (3.2A peak) per channel and allows high-frequency PWM for accurate speed control. The module operates with a logic voltage of 2.7V to 5.5V and supports motor supply voltages up to 15V. Notable features include built-in thermal shutdown and undervoltage protection
10	D6, D9, D10, D5, D3, D11	Output Pins for Dual Motor Driver TB6612FNG: > D9 and D10 – Control AIN1 and AIN2 (direction control for Motor 1) > ~6 (PWM) – Connected to PWMA, used for controlling the speed of Motor 1 > D10 and D5 – Control BIN1 and BIN2 (direction control for Motor 2) > ~5 (PWM) – Connected to PWMB, used for controlling the speed of Motor 2
3	Arduino Nano Slot	Serves as the connection interface for the Arduino Nano microcontroller.
13	D4, D2	Pinout for connecting a 2-pin DIP switch. By configuring the pin as INPUT_PULLUP, the default state will be HIGH, and it will toggle to LOW when the switch is turned ON.
8	VIN, GND	Used to supply power to the circuit. Safe operating voltage ranges from 7V to 12V, with a maximum input limit of 15V.
12	D12	Connected to the tactile switch pin. Configure the pin as INPUT_PULLUP to keep it HIGH by default; it will read LOW when the switch is pressed (ON).
14	RX, TX	Connected to Arduino Nano pins D0 and D1, this interface is used for connecting various Bluetooth and wireless modules such as HC-05, HC-06, or BLE modules.



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9	VIN	Connects a 1N4001 diode to the Arduino Nano's VIN pin to safeguard the circuit against reverse voltage.
15	D2, D4, D13, D12, D7, D8	Primarily designed for connecting ultrasonic sensors, allowing up to three sensors to be connected easily. Additionally, the RX and TX pins can be used to add a fourth ultrasonic sensor if needed.
7	ON/OFF	Used to connect a 3-way switch for turning the circuit board ON or OFF.
6	VIN, GND, 5vExt	Used to connect a voltage regulator module that supplies an external 5V to the circuit board.
5	5vNano	Directly linked to the Arduino Nano's 5V pin. Connect this pin to the 5V power line using a jumper to supply power to all 5V connections from the Nano
5	5vExt	Directly connected to the 5V pin of the external 5V module. Use a jumper to link this pin with the 5VPow line, powering all 5V connections from the external module
5	5vPOW	Directly connected to all 5V pins on the circuit board. You can select either 5vNano or 5vPow as the power source.
4	A0	Pin designated for connecting a passive buzzer. Located beneath the Arduino microcontroller, it helps save valuable circuit board space.

Parts List:

QTY	Part Name
5	3P Terminal Block
3	2P Terminal Block
3	1x3 male header
4	1x4 male header
1	1x4 female header
2	1x8 female header
2	1x15 female header
1	DIP Switch
1	Tactile Switch
3	2P XH connector
1	3 way switch
1	Buzzer
1	1N4001 Diode
4	Nylon Screw and Bolt

Sample Code for Motor Control:

```
void setup() {  
  //M1 pins  
  pinMode(6, OUTPUT); // PWM speed controller (0-255)  
  pinMode(9, OUTPUT); // motor wire connected to D9  
  pinMode(10, OUTPUT); // motor wire connected to D10  
  // M2 pins  
  pinMode(5, OUTPUT); // PWM speed controller (0-255)  
  pinMode(3, OUTPUT); // motor wire connected to D3  
  pinMode(11, OUTPUT); // motor wire connected to D11  
}  
void loop() {  
  // replace numerical value (0-255) to adjust speed  
  forward(100);  
  delay(2000);  
  turnLeft(100);  
  delay(2000);  
  turnRight(100);  
  delay(2000);  
}  
void forward(int speed) {  
  digitalWrite(9, 1);  
  digitalWrite(10, 0);  
  digitalWrite(3, 1);  
  digitalWrite(11, 0);  
  digitalWrite(6, speed);  
  digitalWrite(5, speed);  
}  
void turnLeft(int speed) {  
  digitalWrite(9, 0);  
  digitalWrite(10, 1);  
  digitalWrite(3, 1);  
  digitalWrite(11, 0);  
  analogWrite(6, speed);  
  analogWrite(5, speed);  
}  
void turnRight(int speed) {  
  digitalWrite(9, 1);  
  digitalWrite(10, 0);  
  digitalWrite(3, 0);  
  digitalWrite(11, 1);  
  analogWrite(6, speed);  
  analogWrite(5, speed);  
}
```